

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- e. Bonus problems are not required for credit, but they are worth extra credit.

Problems

Problem 1. (Polynomial Kernel.) Let the feature map be

$$\phi(x) = (1, x, x^2)^\top.$$

- (a.) Write the corresponding kernel $K(x, y) = \phi(x)^\top \phi(y)$.
- (b.) Compute $K(1, 2)$ and $K(0, 3)$.
- (c.) Is this kernel linear, polynomial, or nonlinear? Explain.

Problem 2. (Kernel Matrix Structure.) Let $x_1 = 0$, $x_2 = 1$, $x_3 = 2$ and $k(x, x') = (1 + xx')^2$.

- (a) Compute the kernel matrix $K \in \mathbb{R}^{3 \times 3}$.
- (b) Compute $\text{rank}(K)$.

Problem 3. (Primal–Dual Consistency.) Show that the primal and dual kernel ridge regression predictions are identical: that is, show that

$$\Phi \hat{\theta} = K \hat{\alpha}, \quad \text{where } \hat{\theta} = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top y, \hat{\alpha} = (K + \lambda I)^{-1} y, K = \Phi \Phi^\top.$$

Problem 4. (Centered Kernel Matrix.)

Let K be any $n \times n$ kernel matrix and $H = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$. The centered kernel matrix is $K_c = H K H$.

- (a) Show that $K_c \mathbf{1} = 0$ and $\mathbf{1}^\top K_c = 0^\top$.
- (b) Show that for any vector $y \in \mathbb{R}^n$,

$$y^\top K_c y = (Hy)^\top K (Hy).$$

- (c) Hence conclude that centering the kernel matrix is equivalent to replacing the features $\phi(x_i)$ by their mean-centered versions $\phi(x_i) - \bar{\phi}$, where $\bar{\phi} = \frac{1}{n} \sum_i \phi(x_i)$.

Problem 5. (Eigenvalues of the Hat Matrix.) For kernel ridge regression define $H = K(K + \lambda I)^{-1}$.

- (a) Show that H is symmetric.
- (b) Recall that you can diagonalize symmetric PSD matrices like K as $K = Q\Lambda Q^\top$. Show its eigenvalues lie in $(0, 1)$.

Problem 6. (Gradient Check.) For $J(\alpha) = \|y - K\alpha\|^2 + \lambda\alpha^\top K\alpha$:

- (a) Compute $\nabla_\alpha J(\alpha)$.
- (b) Verify that setting it to zero gives $\alpha = (K + \lambda I)^{-1}y$.

Problem 7. (Early Stopping as Regularization.) Gradient descent on $f(\theta) = \frac{1}{2}\|X\theta - y\|^2$ with step size η and $\theta_0 = 0$ yields

$$\theta_t = \eta \sum_{k=0}^{t-1} (I - \eta X^\top X)^k X^\top y.$$

- (a) Show that

$$\sum_{k=0}^{t-1} R^k = (I - R^t)(I - R)^{-1} = (I - R)^{-1}(I - R^t)$$

- (b) Use the fact in part (a) to show that

$$\theta_t = (I - (I - \eta X^\top X)^t)(X^\top X)^{-1} X^\top y.$$

- (c) Diagonalize $X^\top X = Q\Lambda Q^\top$ and express

$$\theta_t = Q \operatorname{diag}\left(\frac{1 - (1 - \eta\lambda_i)^t}{\lambda_i}\right) Q^\top X^\top y.$$

- (d) Explain how each factor $\frac{1 - (1 - \eta\lambda_i)^t}{\lambda_i}$ acts as a regularization filter.